User Guide: File Download Automation Using Selenium with C# (Automation QA/SDET)

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I. Introduction

This guide explains how to automate file downloads using Selenium with C# and validate the downloaded files by ensuring they are successfully saved in a specified directory. The goal is to automate the process of downloading files using Chrome, dynamically manage download paths, and verify that the downloaded files exist in the target location.

II. Pre-Requisites

To get started, ensure that you have **Visual Studio (2022 or later)** installed along with the **Chrome Browser (latest version)**. Additionally, the following NuGet packages need to be installed in your project:

- **Selenium.WebDriver** – For interacting with the browser and performing web automation.
- **Selenium.Support** – Provides additional support for Selenium, including various utilities.
- **Selenium.WebDriver.ChromeDriver** – Enables the use of ChromeDriver to control Chrome.
- **NUnit** – A unit testing framework to create and run test cases.
- **SeleniumExtras.WaitHelpers** – Provides helper methods to handle dynamic waits effectively.

With these prerequisites configured, you can automate file downloads and validate them efficiently.

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III. Code :

```
using System;
using System.IO;
using NUnit.Framework;
using OpenQA.Selenium;
using OpenQA.Selenium.Chrome;
using OpenQA.Selenium.Support.UI;
using SeleniumExtras.WaitHelpers;

namespace SauceLabsAutomationPOM.Selenium_Tutorials
{
    [TestFixture]
    public class FileDownload
    {
        private IWebDriver driver;
        private static string workingDirectory = Directory.GetCurrentDirectory();
        private static string projectDirectory =
Directory.GetParent(workingDirectory).Parent.Parent.FullName;
        private static string downloadPath = Path.Combine(projectDirectory, "AutomationFiles", "Results",
"Downloads");
        private static string Url = "https://the-internet.herokuapp.com/download";

        [OneTimeSetUp]
        public void SetUp()
        {
            // Create download directory if it doesn't exist
            if (!Directory.Exists(downloadPath))
            {
                Directory.CreateDirectory(downloadPath);
            }

            // Configure Chrome options
            var options = new ChromeOptions();
            options.AddUserProfilePreference("download.default_directory", downloadPath);
            options.AddUserProfilePreference("download.prompt_for_download", false);
            options.AddUserProfilePreference("safebrowsing.enabled", true);

            // Initialize driver
            driver = new ChromeDriver(options);
            driver.Manage().Window.Maximize();
            driver.Manage().Timeouts().ImplicitWait = TimeSpan.FromSeconds(20);
        }

        [OneTimeTearDown]
        public void TearDown()
        {
            if (driver != null)
```

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```
        {
            driver.Quit();
        }
    }

    [Test]
    public void FileDownloadTest1()
    {
        driver.Navigate().GoToUrl(Url);
        WebDriverWait wait = new WebDriverWait(driver, TimeSpan.FromSeconds(20));

        // Download 20MB.bin file
        IWebElement fileLink =
wait.Until(ExpectedConditions.ElementIsVisible(By.XPath("//a[normalize-space()='20MB.bin']")));
        string fileName = fileLink.Text;
        Console.WriteLine($"Downloading File: {fileName}");
        fileLink.Click();

        // Validate download
        string downloadedFilePath = Path.Combine(downloadPath, fileName);
        Assert.IsTrue(WaitForFileToDownload(downloadedFilePath), $"File download failed:
{fileName}");
    }

    [Test]
    public void FileDownloadTest2()
    {
        driver.Navigate().GoToUrl(Url);
        WebDriverWait wait = new WebDriverWait(driver, TimeSpan.FromSeconds(20));

        // Download samplePDF.pdf file
        IWebElement fileLink =
wait.Until(ExpectedConditions.ElementIsVisible(By.XPath("//a[normalize-
space()='samplePDF.pdf']")));
        string fileName = fileLink.Text;
        Console.WriteLine($"Downloading File: {fileName}");
        fileLink.Click();

        // Validate download
        string downloadedFilePath = Path.Combine(downloadPath, fileName);
        Assert.IsTrue(WaitForFileToDownload(downloadedFilePath), $"File download failed:
{fileName}");
    }

    private bool WaitForFileToDownload(string filePath, int timeoutInSeconds = 30)
    {
        for (int i = 0; i < timeoutInSeconds; i++)
        {
            if (File.Exists(filePath))
            {
```

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```
                return true;
            }
            System.Threading.Thread.Sleep(1000);
        }
        return false;
    }
  }
}
```

For entire code checkout my Git Repo & Feel Free to clone it :-

https://github.com/kush1107/SauceLabs_SeleniumCSharpPOM/blob/main/Selenium_Tutorials/FIileDownload_Test.cs

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Happy Testing!